The Asymptotic Equipartition Property

Laws of Large Numbers and Large Deviations in Information Theory

Draft Course Notes

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Notation

Logarithms are base 2, written log, in Lectures 1 through 3, so that entropy is measured in bits and matches Cover and Thomas's own convention in Chapter 3. Large deviation rate functions in Lecture 4 are conventionally stated with the natural logarithm, written ln, so that they arise directly as Legendre transforms of a cumulant generating function $\mathbb{E}[e^{\lambda Y}]$. The lecture states the unit at the point where it switches, and the switch changes no qualitative claim, only the base of the exponent.

Symbol	Meaning
$H(X)$	Shannon entropy of X
$D(p \,\|\, q)$	Kullback–Leibler divergence from q to p
$\mathbb{E}[\cdot]$, $\mathrm{Var}(\cdot)$	expectation, variance
$X_1, X_2, \ldots$	i.i.d. random variables
$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$	sample mean
$\hat{p}_n$	empirical proportion (for a $\{0,1\}$-valued source)
$A_\varepsilon^{(n)}$	typical set with parameter ε, sequences of length n
$B_\delta^{(n)}$	smallest set of sequences of length n with probability $\geq 1-\delta$
$a_n \doteq b_n$	equal to first order in the exponent, $\frac{1}{n}\log\frac{a_n}{b_n} \to 0$

Modes of Convergence and the Weak Law of Large Numbers

0.1 Modes of Convergence

A law of large numbers is a statement that a random quantity, the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ of i.i.d. random variables, converges to a constant, the true mean $\mu = \mathbb{E}[X_1]$, as $n \to \infty$. Since $\bar{X}_n$ is itself random, "converges" needs a definition. Three are in standard use.

Definition 0.1 (Modes of convergence)**.** Let $X_1, X_2, \ldots$ be a sequence of random variables and let X be a random variable, all defined on the same probability space. We say X_n converges to X

1. **in probability** if for every $\varepsilon > 0$, $\mathbb{P}\{|X_n - X| > \varepsilon\} \to 0$ as $n \to \infty$,
2. **in mean square** if $\mathbb{E}[(X_n - X)^2] \to 0$ as $n \to \infty$,
3. **almost surely**, or **with probability 1**, if $\mathbb{P}\{\lim_{n\to\infty} X_n = X\} = 1$.

0.2 Markov's and Chebyshev's Inequalities

Both propositions in the previous section leaned on a single simple fact, an inequality due to Markov that converts a bound on an expectation into a bound on a tail probability. It is the workhorse of the whole subject.

Lemma 0.2 (Markov's inequality)**.** *Let X be a nonnegative random variable and let $t > 0$. Then*

$$\mathbb{P}\{X \geq t\} \leq \frac{\mathbb{E}[X]}{t}.$$

Proof. Since $X \geq 0$,

$$X \geq X \cdot \mathbf{1}\{X \geq t\} \geq t \cdot \mathbf{1}\{X \geq t\}.$$

Taking expectations of both sides, using monotonicity of expectation on the left inequality and linearity on the right,

$$\mathbb{E}[X] \geq \mathbb{E}\big[t \cdot \mathbf{1}\{X \geq t\}\big] = t \cdot \mathbb{P}\{X \geq t\}.$$

Dividing by $t > 0$ gives the claim. □

Markov's inequality only uses that X is nonnegative and has a finite mean. Applied to the squared deviation of a random variable from its mean, it becomes a statement about how concentrated the variable is around its mean, in terms of its variance.

Lemma 0.3 (Chebyshev's inequality)**.** *Let Y be a random variable with mean $\mu = \mathbb{E}[Y]$ and variance $\sigma^2 = \mathrm{Var}(Y) < \infty$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\{|Y - \mu| > \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Let $X = (Y-\mu)^2$, a nonnegative random variable with $\mathbb{E}[X] = \sigma^2$. The event $|Y-\mu| > \varepsilon$ is the same event as $X > \varepsilon^2$, so applying Theorem 0.2 to X with threshold $t = \varepsilon^2$,

$$\mathbb{P}\{|Y - \mu| > \varepsilon\} = \mathbb{P}\{X > \varepsilon^2\} \leq \mathbb{P}\{X \geq \varepsilon^2\} \leq \frac{\mathbb{E}[X]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$ □

This is exactly Problem 3.1 of Cover and Thomas (2006), parts (a) and (b), given here with a complete proof rather than left as an exercise. Part (c) of that same problem is the weak law of large numbers, which is the content of the next section.

0.3 The Weak Law of Large Numbers

Theorem 0.4 (Weak law of large numbers)**.** *Let $Z_1, Z_2, \dots$ be i.i.d. random variables with mean μ and finite variance σ^2. Let $\bar{Z}_n = \frac{1}{n}\sum_{i=1}^{n} Z_i$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\{|\bar{Z}_n - \mu| > \varepsilon\} \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0 \quad \text{as } n \to \infty,$$

so $\bar{Z}_n \to \mu$ in probability.

Proof. By linearity of expectation, $\mathbb{E}[\bar{Z}_n] = \mu$. By independence, the variance of a sum is the sum of the variances, so

$$\mathrm{Var}(\bar{Z}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} Z_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(Z_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Apply Theorem 0.3 to $Y = \bar{Z}_n$, with mean μ and variance σ^2/n:

$$\mathbb{P}\{|\bar{Z}_n - \mu| > \varepsilon\} \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

For fixed $\varepsilon > 0$ and $\sigma^2 < \infty$, the right side tends to 0 as $n \to \infty$, which is exactly the statement that $\bar{Z}_n \to \mu$ in probability. □

0.4 The Asymptotic Equipartition Property

Let $X_1, X_2, \ldots$ be i.i.d. random variables drawn from a probability mass function $p(x)$ on a finite alphabet $\mathcal{X}$. Write $p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i)$ for the probability of observing a particular length n sequence. Consider the random variable

$$Z_i = -\log p(X_i).$$

Since the X_i are i.i.d., and Z_i is a fixed function of X_i alone, the Z_i are i.i.d. as well. Their common expectation is

$$\mathbb{E}[Z_i] = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = H(X),$$

the Shannon entropy of X, by definition. So entropy is nothing more exotic than the expected value of a specific random variable, $-\log p(X)$, sometimes called the information content, or the surprise, of the outcome X. Rare outcomes have small $p(x)$ and large $-\log p(x)$, so they carry a large surprise, and entropy is the average surprise.

Now look at the sample mean of $Z_1, \ldots, Z_n$:

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^{n} Z_i = -\frac{1}{n} \sum_{i=1}^{n} \log p(X_i) = -\frac{1}{n} \log \prod_{i=1}^{n} p(X_i) = -\frac{1}{n} \log p(X_1, \ldots, X_n).$$

This is the quantity Cover and Thomas call the sample entropy of the sequence $X_1, \ldots, X_n$. It is a sample mean of i.i.d. random variables, in the exact sense of Theorem 0.4, so the weak law of large numbers applies to it directly, with $\mu = H(X)$ in place of a generic mean.

0.5 The AEP Theorem

Theorem 0.5 (Asymptotic equipartition property, Cover and Thomas, 2006, Theorem 3.1.1)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables drawn from $p(x)$ on a finite alphabet $\mathcal{X}$. Then*

$$-\frac{1}{n} \log p(X_1, X_2, \ldots, X_n) \longrightarrow H(X) \quad \textit{in probability.}$$

Proof. Let $Z_i = -\log p(X_i)$. Since $\mathcal{X}$ is finite, Z_i takes at most $|\mathcal{X}|$ distinct values, each of them finite because $p(x) > 0$ for every x that actually occurs with positive probability, so Z_i is a bounded random variable. A bounded random variable automatically has finite variance, so $Z_1, Z_2, \ldots$ are i.i.d. with mean $H(X)$ and finite variance, exactly the hypotheses of Theorem 0.4. Applying Theorem 0.4 to $Z_1, \ldots, Z_n$,

$$\bar{Z}_n = -\frac{1}{n} \log p(X_1, \ldots, X_n) \longrightarrow \mathbb{E}[Z_1] = H(X) \quad \text{in probability,}$$

which is the claim. □

0.6 Typical Sets and High-Probability Sets

The asymptotic equipartition property says that $-\frac{1}{n}\log p(X^n)$ concentrates near $H(X)$.

0.7 The Typical Set

Definition 0.6 (Typical set, Cover and Thomas, 2006, Chapter 3.1)**.** Fix $\varepsilon > 0$. The typical set $A_\varepsilon^{(n)}$ with respect to $p(x)$ is the set of sequences $(x_1, \dots, x_n) \in \mathcal{X}^n$ satisfying

$$2^{-n(H(X)+\varepsilon)} \le p(x_1, \dots, x_n) \le 2^{-n(H(X)-\varepsilon)}.$$

A sequence is typical, in this technical sense, exactly when its probability sits within a factor of $2^{n\varepsilon}$ of $2^{-nH(X)}$ in either direction, equivalently when its sample entropy $-\frac{1}{n}\log p(x^n)$ lies within ε of $H(X)$.

Theorem 0.7 (Properties of the typical set, Cover and Thomas, 2006, Theorem 3.1.2)**.** *For every $\varepsilon > 0$:*

1. *If $(x_1, \dots, x_n) \in A_\varepsilon^{(n)}$, then $H(X) - \varepsilon \le -\frac{1}{n}\log p(x_1, \dots, x_n) \le H(X) + \varepsilon$.*
2. *$\mathbb{P}\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$ for n sufficiently large.*
3. *$|A_\varepsilon^{(n)}| \le 2^{n(H(X)+\varepsilon)}$.*
4. *$|A_\varepsilon^{(n)}| \ge (1-\varepsilon)2^{n(H(X)-\varepsilon)}$ for n sufficiently large.*

Proof. Property 1. This is a restatement of Theorem 0.6: divide the defining inequality by $-n$, which reverses both inequalities, so $H(X) + \varepsilon \ge -\frac{1}{n}\log p(x^n) \ge H(X) - \varepsilon$.

Property 2. By Theorem 0.5, $-\frac{1}{n}\log p(X^n) \to H(X)$ in probability, which by Theorem 0.1 means exactly that for every $\varepsilon > 0$,

$$\mathbb{P}\Big\{\big|-\tfrac{1}{n}\log p(X^n) - H(X)\big| > \varepsilon\Big\} \to 0 \quad \text{as } n \to \infty.$$

So for any target $\delta > 0$ there is an n_0 such that for $n \ge n_0$,

$$\mathbb{P}\Big\{\big|-\tfrac{1}{n}\log p(X^n) - H(X)\big| \le \varepsilon\Big\} > 1 - \delta.$$

The event on the left is precisely the event $X^n \in A_\varepsilon^{(n)}$, by Property 1. Setting $\delta = \varepsilon$, which we are free to do since δ was an arbitrary target, gives $\mathbb{P}\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$ for $n \ge n_0$, that is, for n sufficiently large.

Property 3. Since probabilities sum to 1 over the whole alphabet $\mathcal{X}^n$, and $A_\varepsilon^{(n)} \subseteq \mathcal{X}^n$,

$$1 = \sum_{x^n \in \mathcal{X}^n} p(x^n) \ge \sum_{x^n \in A_\varepsilon^{(n)}} p(x^n) \ge \sum_{x^n \in A_\varepsilon^{(n)}} 2^{-n(H(X)+\varepsilon)} = 2^{-n(H(X)+\varepsilon)}|A_\varepsilon^{(n)}|,$$

where the second inequality drops all the non-typical terms, which are nonnegative, and the third uses the lower bound in Theorem 0.6 for every typical sequence. Rearranging, $|A_\varepsilon^{(n)}| \le 2^{n(H(X)+\varepsilon)}$.

Property 4. By Property 2, for n large enough that $\mathbb{P}\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$,

$$1 - \varepsilon < \mathbb{P}\{A_\varepsilon^{(n)}\} = \sum_{x^n \in A_\varepsilon^{(n)}} p(x^n) \le \sum_{x^n \in A_\varepsilon^{(n)}} 2^{-n(H(X)-\varepsilon)} = 2^{-n(H(X)-\varepsilon)}|A_\varepsilon^{(n)}|,$$

using the upper bound in Theorem 0.6 for every typical sequence. Rearranging, $|A_\varepsilon^{(n)}| \ge (1-\varepsilon)2^{n(H(X)-\varepsilon)}$. □

0.8 Consequence: A Short Description for Almost Every Sequence

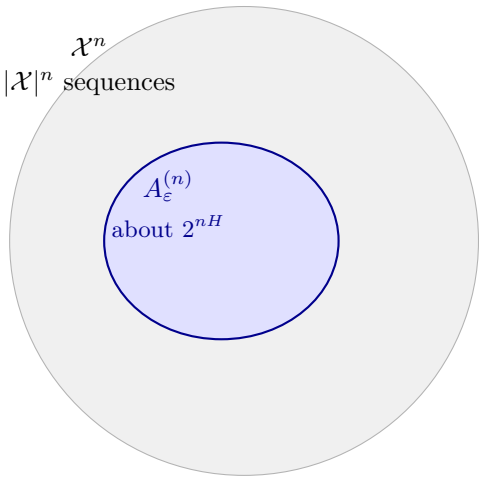


Figure 0.1: The typical set $A_\varepsilon^{(n)}$ (solid blue)

Theorem 0.7 lets us build a short description for a sequence drawn from $p(x)$, without knowing anything else about the source. List the sequences of $A_\varepsilon^{(n)}$ in some fixed order, say lexicographic order. There are at most $2^{n(H+\varepsilon)}$ of them by Property 3, so an index into this list takes at most $\lceil n(H+\varepsilon)\rceil \le n(H+\varepsilon)+1$ bits. Do the same for the complement, $(A_\varepsilon^{(n)})^c$, which has at most $|\mathcal{X}|^n$ elements trivially, so an index into it takes at most $n\log|\mathcal{X}|+1$ bits. Prefix a single flag bit, 0 for a codeword drawn from the typical list and 1 for the atypical list, ahead of each index. This gives a code, one to one and instantly decodable since the first bit alone tells the decoder which list, and hence which length, to expect, with codeword length

$$l(x^n) \le \begin{cases} n(H+\varepsilon)+2 & \text{if } x^n \in A_\varepsilon^{(n)}, \\ n\log|\mathcal{X}|+2 & \text{if } x^n \notin A_\varepsilon^{(n)}. \end{cases}$$

Theorem 0.8 (Short expected description length, Cover and Thomas, 2006, Theorem 3.2.1)**.** *Let X^n be i.i.d. $\sim p(x)$ and let $\varepsilon > 0$. There is a code mapping sequences x^n to binary strings, one to one, with*

$$\mathbb{E}\left[\frac{1}{n}l(X^n)\right] \le H(X)+\varepsilon$$

for n sufficiently large.

Proof. Using the two-case bound on $l(x^n)$ above,

$$\begin{aligned}\mathbb{E}[l(X^n)] &= \sum_{x^n\in A_\varepsilon^{(n)}} p(x^n)l(x^n) + \sum_{x^n\notin A_\varepsilon^{(n)}} p(x^n)l(x^n) \\ &\le \mathbb{P}\{A_\varepsilon^{(n)}\}\big(n(H+\varepsilon)+2\big) + \mathbb{P}\{(A_\varepsilon^{(n)})^c\}\big(n\log|\mathcal{X}|+2\big) \\ &= \mathbb{P}\{A_\varepsilon^{(n)}\}\cdot n(H+\varepsilon) + \mathbb{P}\{(A_\varepsilon^{(n)})^c\}\cdot n\log|\mathcal{X}| + 2\Big(\mathbb{P}\{A_\varepsilon^{(n)}\} + \mathbb{P}\{(A_\varepsilon^{(n)})^c\}\Big).\end{aligned}$$

The term $\mathbb{P}\{A_\varepsilon^{(n)}\} + \mathbb{P}\{(A_\varepsilon^{(n)})^c\}$ equals 1 exactly, since $A_\varepsilon^{(n)}$ and its complement partition $\mathcal{X}^n$, so the last term is exactly 2, with no approximation. Bounding the two remaining probabilities crudely, $\mathbb{P}\{A_\varepsilon^{(n)}\} \le 1$ and, by Theorem 0.7 Property 2, $\mathbb{P}\{(A_\varepsilon^{(n)})^c\} < \varepsilon$ for n sufficiently large,

$$\mathbb{E}[l(X^n)] \le n(H+\varepsilon) + \varepsilon\, n\log|\mathcal{X}| + 2 = n(H+\varepsilon'), \qquad \varepsilon' = \varepsilon + \varepsilon\log|\mathcal{X}| + \frac{2}{n}.$$

Dividing by n gives $\mathbb{E}[\frac{1}{n}l(X^n)] \leq H + \varepsilon'$. Choosing ε small makes $\varepsilon + \varepsilon \log |\mathcal{X}|$ small, and then choosing n large makes $2/n$ small, so ε' can be made as small as we like, which is the content of the theorem, with ε' relabeled ε. $\square$

0.9 Large Deviations and the Exponential Rate of the AEP

The asymptotic equipartition property says $-\frac{1}{n}\log p(X^n)$ converges to $H(X)$ in probability, and Chebyshev bound turns this into an explicit tail bound that shrinks like $1/n$.

The probability of a deviation from $H(X)$ does not shrink like $1/n$, it shrinks exponentially in n, and the exponential rate is itself a Kullback-Leibler divergence. This is the theory of large deviations, and Kullback-Leibler divergence appearing as its rate function, for sums of i.i.d. random variables, is not a coincidence adjacent to information theory. It is one of the clearest instances of entropy appearing inside classical probability theory on its own terms, with no information theoretic motivation required to derive it.

Because rate functions are conventionally written with the natural logarithm, this lecture switches from $\log = \log_2$, used in Lectures 1 through 3, to ln, the natural logarithm, and states so at each place the base matters.

Learning objectives

- Prove the Chernoff bound for the tail probability of a sum of i.i.d. random variables.
- Define the large deviation rate function as a Legendre transform, and state Cramér's theorem.
- Derive, in closed form, that the rate function for the sample proportion of a Bernoulli source is the binary Kullback-Leibler divergence.
- Prove a companion result, that the log likelihood ratio between two hypotheses concentrates at rate $D(p\|q)$.
- Compare a Chebyshev bound and a Chernoff bound numerically on a concrete example, and see the exponential rate directly.

0.10 The Chernoff Bound

Theorem 0.9 (Chernoff bound)**.** *Let $Y_1, \ldots, Y_n$ be i.i.d. real valued random variables with cumulant generating function $\Lambda(\lambda) = \ln \mathbb{E}[e^{\lambda Y_1}]$, finite for λ in some interval containing 0. Let $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i$ and let $a > \mathbb{E}[Y_1]$. Then for every $\lambda \geq 0$ for which $\Lambda(\lambda)$ is finite,*

$$\mathbb{P}\{\bar{Y}_n \geq a\} \leq e^{-n(\lambda a - \Lambda(\lambda))},$$

and consequently

$$\mathbb{P}\{\bar{Y}_n \geq a\} \leq e^{-n\Lambda^*(a)}, \qquad \Lambda^*(a) := \sup_{\lambda \geq 0}\big(\lambda a - \Lambda(\lambda)\big).$$

Proof. Fix $\lambda \geq 0$. Since $x \mapsto e^{\lambda x}$ is nondecreasing for $\lambda \geq 0$, the event $\bar{Y}_n \geq a$ is the same event as $e^{\lambda n \bar{Y}_n} \geq e^{\lambda n a}$, and $e^{\lambda n \bar{Y}_n} = e^{\lambda \sum_i Y_i} = \prod_i e^{\lambda Y_i}$ is a nonnegative random variable, so Theorem 0.2 applies to it directly:

$$\mathbb{P}\{\bar{Y}_n \geq a\} = \mathbb{P}\Big\{\prod_i e^{\lambda Y_i} \geq e^{\lambda n a}\Big\} \leq \frac{\mathbb{E}\big[\prod_i e^{\lambda Y_i}\big]}{e^{\lambda n a}}.$$

Since $Y_1, \ldots, Y_n$ are independent, $\mathbb{E}\big[\prod_i e^{\lambda Y_i}\big] = \prod_i \mathbb{E}[e^{\lambda Y_i}] = \big(\mathbb{E}[e^{\lambda Y_1}]\big)^n$ by identical distribution, so

$$\mathbb{P}\{\bar{Y}_n \geq a\} \leq \frac{\big(\mathbb{E}[e^{\lambda Y_1}]\big)^n}{e^{\lambda n a}} = \frac{e^{n\Lambda(\lambda)}}{e^{\lambda n a}} = e^{-n(\lambda a - \Lambda(\lambda))}.$$

This holds for every $\lambda \geq 0$ in the domain of Λ, and the left side does not depend on λ, so it holds in particular for the λ that makes the right side smallest, which is the λ that makes $\lambda a - \Lambda(\lambda)$ largest, giving the bound $e^{-n\Lambda^*(a)}$. □

Connection to Probability Theory

The proof uses exactly one tool, Markov's inequality (Theorem 0.2), applied not to $\bar{Y}_n$ itself but to $e^{\lambda n \bar{Y}_n}$, an exponential tilt of it. Chebyshev's inequality, used throughout Lectures 1 through 3, is the special case of this same idea applied to a different, non-exponential function, $(\bar{Y}_n - \mu)^2$. Exponential tilting turns out to capture the true rate of decay, quadratic tilting does not, and the difference between the two is the entire content of this lecture.

0.11 The Rate Function and Cramér's Theorem

Definition 0.10 (Large deviation rate function)**.** For a random variable Y with cumulant generating function $\Lambda(\lambda) = \ln \mathbb{E}[e^{\lambda Y}]$, the rate function is the Legendre transform

$$\Lambda^*(a) = \sup_{\lambda \geq 0} (\lambda a - \Lambda(\lambda)), \qquad a > \mathbb{E}[Y].$$

Theorem 0.9 proves half of the theory of large deviations for i.i.d. sums, an upper bound on the tail probability, using nothing more than Markov's inequality. The other half, that this exponential rate is not just an upper bound but the true rate, is a genuinely deeper theorem.

Theorem 0.11 (Cramér's theorem, upper bound direction proved, Cramér, 1938)**.** *Under the hypotheses of Theorem 0.9, and mild regularity conditions on* Λ *satisfied whenever* Y_1 *is bounded or has a finite moment generating function in a neighborhood of* 0*,*

$$\lim_{n \to \infty} \frac{1}{n} \ln \mathbb{P}\{\bar{Y}_n \geq a\} = -\Lambda^*(a).$$

We proved the inequality $\frac{1}{n} \ln \mathbb{P}\{\bar{Y}_n \geq a\} \leq -\Lambda^*(a)$ in Theorem 0.9. The matching inequality in the other direction, that the probability is not asymptotically smaller than $e^{-n\Lambda^*(a)}$, is the hard half of Cramér's theorem, and we do not prove it here. The usual proof constructs a new probability measure, the exponentially tilted or Esscher transformed measure at the optimal λ, under which a becomes the typical value rather than a rare one, and applies a local, more delicate, version of the law of large numbers under that tilted measure. See Dembo and Zeitouni (1998, Chapter 2) for a complete modern proof, or Cramér's own original paper (Cramér, 1938) for the first proof, in the special case of sums of i.i.d. real random variables.

Key Idea

The pattern in this lecture repeats the pattern of Lecture 1 exactly. There, the weak law of large numbers was an easy one line consequence of Chebyshev's inequality, and the strong law needed genuinely more machinery, which we cited. Here, the Chernoff upper bound is an easy one line consequence of Markov's inequality, and the matching lower bound, the real content of Cramér's theorem, needs genuinely more machinery, which we again cite rather than prove. The same division of labor, an elementary inequality proves an upper bound easily, and a matching lower bound is the hard direction, shows up throughout probability theory.

0.12 The Bernoulli Rate Function Is a Relative Entropy

We now compute $\Lambda^*(a)$ explicitly for a source that has appeared throughout these notes.

Proposition 0.12 (Bernoulli rate function)**.** *Let $Y_i = \mathbb{1}\{X_i = 1\}$ where $X_1, X_2, \ldots$ are i.i.d. with $\mathbb{P}\{X_i = 1\} = p$, $\mathbb{P}\{X_i = 0\} = q = 1 - p$, so $\bar{Y}_n = \hat{p}_n$, the empirical proportion of ones. For $a \in (p, 1)$,*

$$\Lambda^*(a) = a \ln \frac{a}{p} + (1 - a) \ln \frac{1 - a}{1 - p} = D(a \| p),$$

the binary Kullback-Leibler divergence, in nats.

Proof. The cumulant generating function of Y_1 is

$$\Lambda(\lambda) = \ln \mathbb{E}[e^{\lambda Y_1}] = \ln(q + pe^{\lambda}).$$

Differentiating $\lambda a - \Lambda(\lambda)$ with respect to λ and setting the derivative to 0,

$$a - \frac{pe^{\lambda}}{q + pe^{\lambda}} = 0 \quad \Longrightarrow \quad a(q + pe^{\lambda}) = pe^{\lambda} \quad \Longrightarrow \quad e^{\lambda} = \frac{aq}{p(1 - a)}.$$

For $a \in (p, 1)$ this gives $e^{\lambda} > 1$, so the maximizing $\lambda^* = \ln \frac{aq}{p(1-a)}$ is indeed positive, inside the domain $\lambda \geq 0$ of Theorem 0.10. At this λ^*,

$$q + pe^{\lambda^*} = q + p \cdot \frac{aq}{p(1 - a)} = q \left(1 + \frac{a}{1 - a}\right) = \frac{q}{1 - a},$$

so

$$\Lambda(\lambda^*) = \ln \frac{q}{1 - a} = \ln q - \ln(1 - a).$$

And

$$\lambda^* a = a \ln \frac{aq}{p(1 - a)} = a \ln a + a \ln q - a \ln p - a \ln(1 - a).$$

Subtracting,

$$\Lambda^*(a) = \lambda^* a - \Lambda(\lambda^*) = a \ln a - a \ln p - a \ln(1 - a) + a \ln q - \ln q + \ln(1 - a).$$

Collect the terms with $a \ln a - a \ln p = a \ln(a/p)$ on one side, and group the rest,

$$-a \ln(1 - a) + \ln(1 - a) + a \ln q - \ln q = (1 - a) \ln(1 - a) - (1 - a) \ln q = (1 - a) \ln \frac{1 - a}{q},$$

using $q = 1 - p$. Combining,

$$\Lambda^*(a) = a \ln \frac{a}{p} + (1 - a) \ln \frac{1 - a}{1 - p} = D(a \| p). \qquad \square$$

We checked Theorem 0.12 two ways before writing it down. The algebra above was carried out symbolically, and separately, for several numerical pairs (a, p), the supremum defining $\Lambda^*(a)$ was found by direct numerical optimization over λ and compared against $D(a\|p)$ computed independently, agreeing to floating point precision in every case tried.

Corollary 0.13 (Exponential concentration of the sample proportion)**.** *For $a \in (p, 1)$,*

$$\mathbb{P}\{\hat{p}_n \geq a\} \leq e^{-nD(a\|p)},$$

and this bound is tight to first order in the exponent, by Theorem 0.11.

Connection to Probability Theory

This is the sharpened form of the law of large numbers that Lecture 1 promised. Theorem 0.4 said $\hat{p}_n \to p$ in probability with a bound of order $1/n$ on the tail probability, from Chebyshev's inequality. Theorem 0.13 replaces that $1/n$ bound with $e^{-nD(a\|p)}$, an exponentially smaller quantity for large n, and relative entropy, the same quantity that measures the excess coding cost of using the wrong distribution in Lecture 2's cross entropy discussion, is exactly the rate at which this probability vanishes. Since the sample entropy $-\frac{1}{n}\log p(X^n)$ of Lectures 2 and 3 is an increasing or decreasing affine function of $\hat{p}_n$ depending on the sign of $\log(q/p)$, this also gives the asymptotic equipartition property itself an explicit exponential rate, answering the question left open since Theorem 0.5.

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